

Integration — Lesson 1 Test: Adding Up Little Pieces

10 questions · show your work in the box · calculators allowed · every answer needs UNITS

Name: _____ Date: _____ Score: _____ / 10

1. A train travels at a steady 80 miles per hour for 3 hours. How far does it go? 1 pt

2. A faucet fills a tub at 6 liters per minute for 12 minutes. How much water is in the tub? 1 pt

3. A turtle walks 36 meters in 9 minutes at a perfectly steady pace. What is its speed in meters per minute? 1 pt

4. You bike at 10 mph for 2 hours, then at 6 mph for 3 hours. What is your total distance? 1 pt

5. A truck drives 40 mph for 1 hour, 60 mph for 2 hours, and 30 mph for half an hour. Find the total distance. 1 pt

6. A speed graph shows one flat step: 25 km/h from $t = 0$ to $t = 4$ hours. What is the area of the rectangle under the step, and what does that area mean in real life? 1 pt

7. A speed graph is a staircase: 6 m/s from $t = 0$ to $t = 3$ seconds, then 9 m/s from $t = 3$ to $t = 8$ seconds. Use rectangle areas to find the total distance. (Careful with the width of the second rectangle!) 1 pt

8. A jet flies at 600 miles per hour for 45 minutes. How far does it fly? (Watch the units.) 1 pt

9. Water pours INTO a tank at 12 liters per minute while a drain removes 5 liters per minute. Both run for 8 minutes. How much water does the tank gain?

1 pt

10. A runner wants to cover exactly 26 miles. She runs 8 mph for 2 hours, then 5 mph for 1 hour. She finishes the race at 5 mph. How many more hours must she run?

1 pt
